Lecture 4

The Legendre transform and Hamilton's equations

Objectives. *Lecture 3 gave the first answer: a motion makes the action stationary, and its momentum is $p = \partial\mathcal{L}/\partial\dot{q}$. So far a state has been a position with a velocity. This lecture asks when the velocity can be traded for the momentum. Momentum is a covector and transforms differently from velocity. The Legendre transform makes the change possible when the Lagrangian is hyperregular. Lagrange's n second-order equations then become Hamilton's $2n$ first-order equations on the space of positions and momenta, the* phase space*, with an action principle of their own. This takes us from the Lagrange description to the Hamilton description used in Part II.*

By the end of this lecture you will be able to:

- Transform momenta as covectors under a change of configuration coordinates.
- Construct the Hamiltonian of a hyperregular Lagrangian by the Legendre transform.
- Derive Hamilton's equations from Lagrange's equations.
- Formulate the phase-space action with its endpoint conditions.

4.1 Momentum is a covector

Throughout, q ranges over an open set $U \subseteq \mathbb{R}^n$, a patch of generalised coordinates on a configuration space Q or $\mathbb{R}^n$ itself, and $\mathcal{L}(q, \dot{q}, t)$ is a C^2 Lagrangian with momentum $p = \partial\mathcal{L}/\partial\dot{q}$ (Definition 3.9). Momentum has n components, like the velocity. The two transform differently.

Let $\tilde{q} = \varphi(q)$ be other generalised coordinates, with φ a C^2 bijection onto an open set $\tilde{U}$ with invertible derivative (the transition map of Lemma 2.5). Velocities transform by $\dot{\tilde{q}} = D\varphi(q)\,\dot{q}$ (Equation (2.4)). Suppose the same system is described in the new coordinates by a Lagrangian $\tilde{\mathcal{L}}$ with the same value at the same state,

$$\mathcal{L}(q, \dot{q}, t) = \tilde{\mathcal{L}}\big(\varphi(q), D\varphi(q)\,\dot{q}, t\big), \tag{4.1}$$

as happens for $T - V$. This is also the transformation rule of Theorem 3.6, with the roles of the coordinates exchanged. Write $\tilde{p} = \partial\tilde{\mathcal{L}}/\partial\dot{\tilde{q}}$.

Proposition 4.1 (The covector law)**.** *At corresponding states,*

$$p = D\varphi(q)^{\mathsf{T}}\,\tilde{p}, \qquad \text{that is} \qquad \tilde{p} = D\varphi(q)^{-\mathsf{T}}\,p, \tag{4.2}$$

and the pairing is invariant: $\tilde{p}\cdot\dot{\tilde{q}} = p\cdot\dot{q}$.

Proof. Differentiate (4.1) in $\dot{q}_j$. The kth velocity slot of $\tilde{\mathcal{L}}$ is $\sum_j (\partial\varphi_k/\partial q_j)\dot{q}_j$, so the chain rule gives $p_j = \sum_k \tilde{p}_k\, \partial\varphi_k/\partial q_j$, which is (4.2). Then $\tilde{p} \cdot D\varphi\, \dot{q} = (D\varphi^{\mathsf{T}}\tilde{p}) \cdot \dot{q} = p \cdot \dot{q}$. ∎

For example, take polar coordinates $\tilde{q} = (\rho, \vartheta)$ in the plane, with $x = \rho\cos\vartheta$ and $y = \rho\sin\vartheta$. The columns of $D\varphi^{-1}$ are $\partial(x,y)/\partial\rho$ and $\partial(x,y)/\partial\vartheta$, so (4.2) gives

$$p_\rho = \frac{x p_x + y p_y}{\rho}, \qquad p_\vartheta = x p_y - y p_x.$$

The second is the angular momentum of § 3.4.

Velocities are multiplied by $D\varphi$, and momenta by $D\varphi^{-\mathsf{T}}$. The two agree only when $D\varphi$ is orthogonal. At a fixed configuration the velocities form a vector space $V \cong \mathbb{R}^n$, and a momentum p defines the linear function $v \mapsto p \cdot v$ on it: an element of the *dual space* V^*. When the components of vectors change by a matrix A, the components of dual vectors change by $A^{-\mathsf{T}}$, preserving the pairing. Proposition 4.1 gives this rule with $A = D\varphi(q)$. Thus *a momentum is a covector*, a linear function of velocities, and p_j is its value on the velocity $\dot{q} = e_j$.

The picture

A covector $p \neq 0$ is drawn as the stack of parallel, equally spaced lines (hyperplanes, for $n > 2$) $\{v : p \cdot v = k\}$, $k \in \mathbb{Z}$. The number $p \cdot v$ is the signed number of lines that the arrow v, drawn from the origin, crosses, counting the last fraction. A large p has closely packed lines. A change of coordinates carries the arrow and the stack along together, so the count cannot change: that is the invariance of the pairing (Figure 4.1).

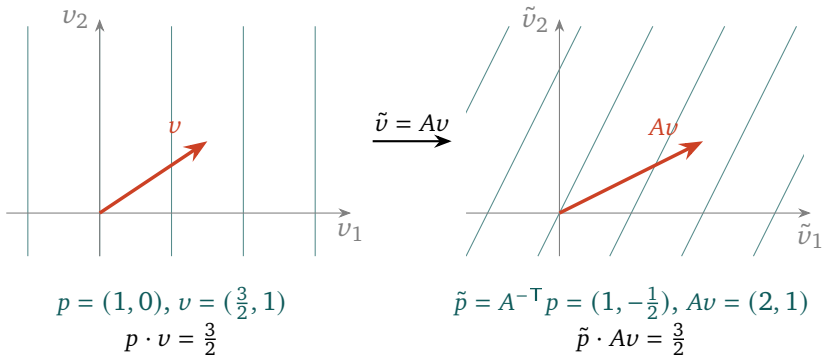


Figure 4.1: A covector as a stack of lines, crossed by a vector, in the space of velocities at one configuration. The coordinate change $\tilde{q} = Aq$ acts on velocities by $\tilde{v} = Av$. The shear $A = \left(\begin{smallmatrix} 1 & 1/2 \\ 0 & 1 \end{smallmatrix}\right)$ tilts the lines $v_1 = k$ into $\tilde{v}_1 - \frac{1}{2}\tilde{v}_2 = k$ and moves the arrow; the arrow still crosses one and a half lines. The shear changes the first component of the vector and the second component of the covector.

Phase space

After replacing velocity by momentum, a state is a configuration together with a covector. In a patch, a point $(q, p) \in U \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$, and on a change of patch

$$(q, p) \longmapsto \big(\varphi(q),\, D\varphi(q)^{-\mathsf{T}} p\big). \tag{4.3}$$

This is the *phase space*. In Lecture 1 phase space was the space of positions and velocities; from now on it is the space of positions and momenta. Over the whole of Q the set of such pairs is called the *cotangent bundle* T^*Q; we work in one patch.

4.2 The Legendre transform of a convex function

At fixed q and t the map $\dot q \mapsto \partial L/\partial \dot q$ is the gradient map of one function. When can a gradient map be inverted?

Theorem 4.2 (The Legendre transform). *Let $U \subseteq \mathbb{R}^n$ be open and convex, and let $f \in C^2(U)$ have a positive-definite Hessian $D^2 f$ at every point.*

1. *The* Legendre map $x \mapsto \nabla f(x)$ *is injective on U, its image U^* is open, and its inverse $p \mapsto x(p)$ is C^1. Define the* Legendre transform
$$g(p) = p \cdot x(p) - f\big(x(p)\big), \qquad p \in U^*. \tag{4.4}$$
2. *$g \in C^2(U^*)$, $\nabla g(p) = x(p)$, and $D^2 g(p) = \big(D^2 f(x(p))\big)^{-1}$ is positive definite.*
3. *The transform is* involutive*: the Legendre map of g is $p \mapsto x(p)$, a bijection of U^* onto U, and the transform of g is f.*
4. Young's inequality: *$p \cdot x \le f(x) + g(p)$ for all $x \in U$, $p \in U^*$, with equality if and only if $p = \nabla f(x)$. Hence $g(p) = \max_{x \in U}\big(p \cdot x - f(x)\big)$.*

Proof. (1) For $x \neq y$ in U the segment between them lies in U, so
$$(\nabla f(y) - \nabla f(x)) \cdot (y - x) = \int_0^1 (y-x)^{\mathsf T} D^2 f(x + s(y-x))(y-x)\, \mathrm{d}s > 0.$$
The derivative $D^2 f$ of ∇f is invertible, so by the inverse function theorem ∇f has C^1 local inverses on open sets; hence U^* is open and the global inverse is C^1.

(2) By the chain rule, $\nabla g = x + Dx^{\mathsf T}\big(p - \nabla f(x(p))\big) = x(p)$. So ∇g is C^1, and $D^2 g = Dx = (D^2 f)^{-1}$, the derivative of an inverse map.

(3) $\nabla g = x(\cdot)$ has inverse ∇f, and at $p = \nabla f(x)$, $x \cdot p - g(p) = f(x)$ by (4.4).

(4) Fix p, let $x_p = x(p)$, $h(x) = p \cdot x - f(x)$ and $u(s) = h(x_p + s(x - x_p))$ for $0 \le s \le 1$. Then $u'(0) = (p - \nabla f(x_p)) \cdot (x - x_p) = 0$ and $u'' < 0$ if $x \neq x_p$, so $h(x) = u(1) < u(0) = g(p)$ unless $x = x_p$. ∎

The pictures

Take $f(x) = \cosh x - 1$. Its Legendre map is $p = f'(x) = \sinh x$, with inverse $x(p) = \operatorname{arsinh} p$. Substituting in (4.4), and using $\cosh(\operatorname{arsinh} p) = \sqrt{1+p^2}$, we obtain
$$g(p) = p \operatorname{arsinh} p - \sqrt{1+p^2} + 1.$$
By Young's inequality in Theorem 4.2, $g(p)$ is the *largest vertical gap* between the line px and the graph of f. By involutivity, the graph is the *envelope* of the tangents $y = px - g(p)$ (Figure 4.2).

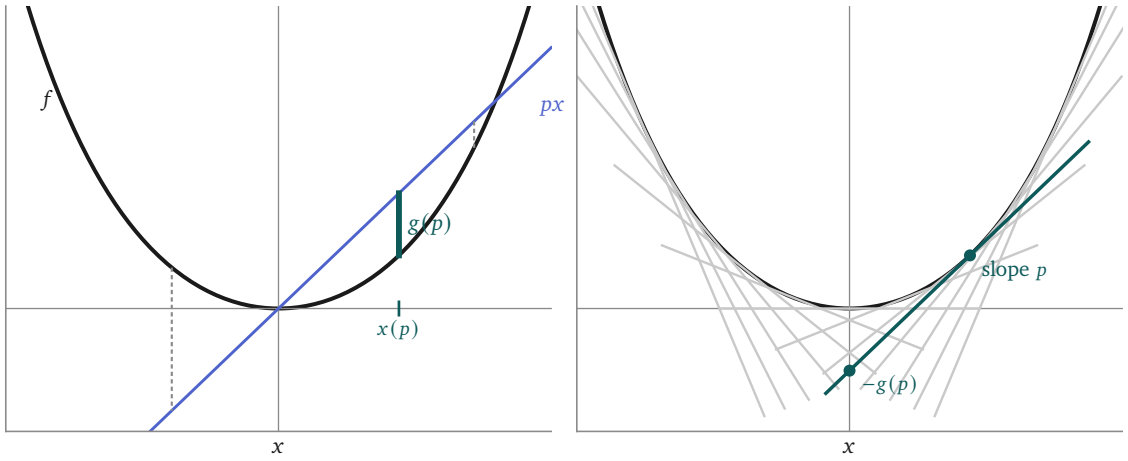


Figure 4.2: The function $f(x) = \cosh x - 1$. Left, the largest gap: for $p = 1.2$ the gap $px - f(x)$ is largest at $x(p)$, where it equals $g(p)$. Right, the envelope: the tangents $y = px - g(p)$, $p = -3, -2.5, \ldots, 3$.

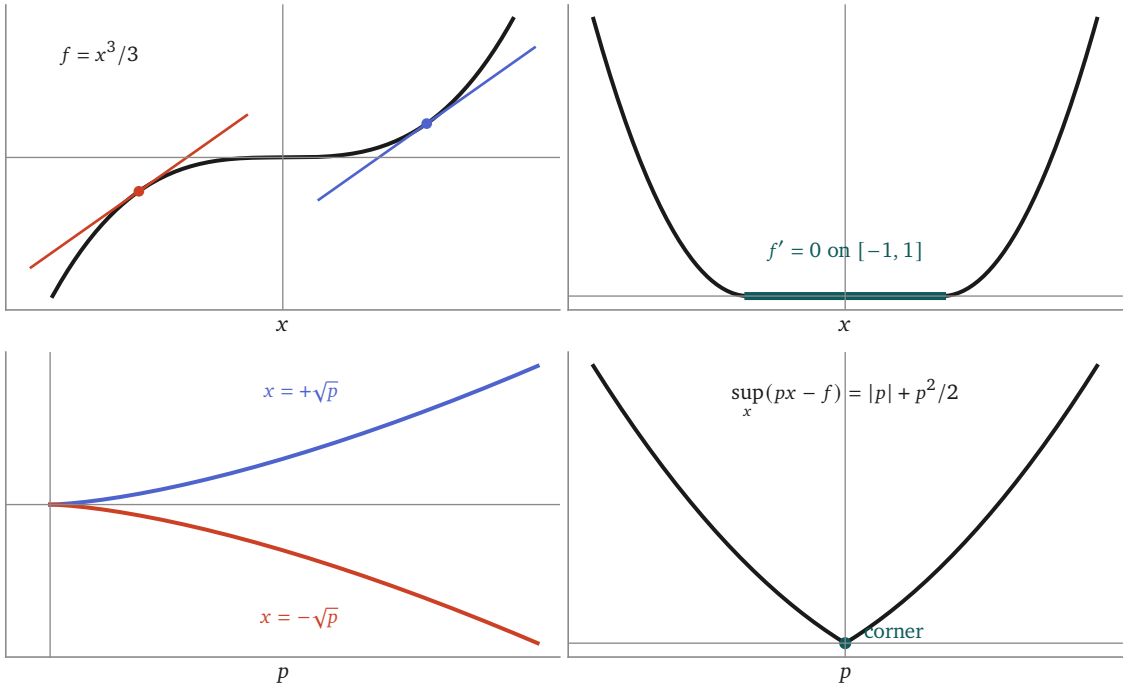


Figure 4.3: Left, an inflection point: two tangents of slope p (top), and two branches of $px - f(x)$ (bottom). Right, a flat piece (top): here $\sup_x(px - f) = |p| + p^2/2$, which has a corner at $p = 0$ (bottom).

Failure

Without a positive-definite Hessian the Legendre map need not be injective (Figure 4.3). For $f = x^3/3$ it folds at the inflection point: each $p > 0$ is x^2 at $x = \pm\sqrt{p}$, where $px - f(x) = \pm\frac{2}{3}p^{3/2}$. For the C^1 function $f = \frac{1}{2}\max(|x| - 1, 0)^2$ it is 0 on the whole flat piece $[-1, 1]$.

4.3 The Legendre transform of a Lagrangian

As in Lecture 3, a Lagrangian is a C^2 function on $U \times \mathbb{R}^n \times I$, defined for all velocities.

Definition 4.3 (Fibre derivative; regular and hyperregular)**.** The *fibre* over (q,t) is the set of velocities at that (q,t). The *fibre derivative* of $\mathcal{L}$ is

$$\mathbb{F}\mathcal{L}(q,\dot q,t) = \Big(q,\ \frac{\partial \mathcal{L}}{\partial \dot q}(q,\dot q,t),\ t\Big),$$

and $W = \partial^2\mathcal{L}/\partial\dot q\,\partial\dot q$ is the $n\times n$ Hessian in the velocities. The Lagrangian $\mathcal{L}$ is *regular* if $\det W \neq 0$ everywhere, and *hyperregular* if it is regular and $\mathbb{F}\mathcal{L}$ is a bijection of $U\times\mathbb{R}^n\times I$ onto itself: at every (q,t) every momentum occurs exactly once.

Bijectivity alone is not enough. The Lagrangian $\frac14\dot q^4$ gives $p = \dot q^3$, whose inverse is not differentiable at 0.

At fixed (q,t), $\mathbb{F}\mathcal{L}$ is the Legendre map of $\mathcal{L}(q,\cdot,t)$. Regularity allows us to solve for the accelerations. Along a C^2 path the Euler–Lagrange equations (3.4) read

$$W\,\ddot q = \frac{\partial \mathcal{L}}{\partial q} - \frac{\partial^2 \mathcal{L}}{\partial \dot q\,\partial q}\,\dot q - \frac{\partial^2 \mathcal{L}}{\partial \dot q\,\partial t}. \tag{4.5}$$

If $\mathcal{L}$ is regular, this can be solved for $\ddot q$: a first-order system for the state $(q,\dot q)$ of Lecture 1. Where W is singular, $\mathcal{L}$ does not determine the accelerations.

Regularity is not enough for a global change to momentum coordinates. The Lagrangian $\mathcal{L} = e^{\dot q}$ has $W = e^{\dot q} > 0$ and an injective fibre derivative $\mathbb{F}\mathcal{L}$, but its momenta are only $p > 0$. The Hamiltonian $H = p\ln p - p$ would therefore be defined only on $p > 0$. Hyperregularity excludes this restriction on the momentum domain.

Lemma 4.4. *The derivative of $\mathbb{F}\mathcal{L}$ has determinant $\det W$. If $\mathcal{L}$ is hyperregular, the inverse of $\mathbb{F}\mathcal{L}$ is C^1.*

Proof. In the variables $(q,\dot q,t)$ the derivative is block lower triangular, with diagonal blocks I, W, 1. The rest is the inverse function theorem, as in the inverse-map argument of Theorem 4.2. ∎

From here on, let $\mathcal{L}$ be hyperregular, and write the inverse as $(q,p,t)\mapsto(q,v(q,p,t),t)$: v is the velocity with $\partial\mathcal{L}/\partial\dot q(q,v,t) = p$. The *Hamiltonian* is

$$H(q,p,t) = p\cdot v(q,p,t) - \mathcal{L}\big(q,v(q,p,t),t\big), \qquad (q,p,t)\in U\times\mathbb{R}^n\times I. \tag{4.6}$$

At $p = \partial\mathcal{L}/\partial\dot q$ its value is $p\cdot\dot q - \mathcal{L} = E_{\mathcal{L}}$: *the Hamiltonian is the energy function of Theorem 3.15, written in momenta.*

Theorem 4.5 (Euler–Lagrange ⇔ Hamilton)**.** *Under these hypotheses:*

1. *$H \in C^2(U\times\mathbb{R}^n\times I)$, and at corresponding points $(q,p,t) = \mathbb{F}\mathcal{L}(q,v,t)$,*

$$\frac{\partial H}{\partial p} = v, \qquad \frac{\partial H}{\partial q} = -\frac{\partial \mathcal{L}}{\partial q}, \qquad \frac{\partial H}{\partial t} = -\frac{\partial \mathcal{L}}{\partial t}. \tag{4.7}$$

2. *A C^2 path $q(t)$ in U satisfies the Euler–Lagrange equations if and only if $(q(t),p(t))$, with $p = \partial\mathcal{L}/\partial\dot q(q,\dot q,t)$, satisfies* Hamilton's equations

$$\dot q = \frac{\partial H}{\partial p}, \qquad \dot p = -\frac{\partial H}{\partial q}. \tag{4.8}$$

3. *Every C^1 solution of (4.8) arises in this way.*

Proof. (1) v is C^1 by Lemma 4.4. Differentiate (4.6) by the chain rule:

$$\frac{\partial H}{\partial p_j} = v_j + \sum_k \Big(p_k - \frac{\partial \mathcal{L}}{\partial \dot q_k}\Big)\frac{\partial v_k}{\partial p_j}, \qquad \frac{\partial H}{\partial q_j} = \sum_k \Big(p_k - \frac{\partial \mathcal{L}}{\partial \dot q_k}\Big)\frac{\partial v_k}{\partial q_j} - \frac{\partial \mathcal{L}}{\partial q_j},$$

with $\mathcal{L}$ and its derivatives at (q, v, t). The brackets vanish by the definition of v; the same computation in t gives the third formula. The right sides of (4.7) are C^1, so H is C^2.

(2) Along the path, $(q, p, t) = \mathbb{F}\mathcal{L}(q, \dot q, t)$, so $v(q, p, t) = \dot q$, and the first equation of (4.8) holds by the derivative identities (4.7), whether or not q is a motion. By the same identities, $-\partial H/\partial q(q, p, t) = \partial \mathcal{L}/\partial q(q, \dot q, t)$, so the second equation reads $\dot p = \partial \mathcal{L}/\partial q$, which is (3.4).

(3) If (q, p) solves (4.8), then $\dot q = v(q, p, t)$ is C^1, so q is C^2, and $\mathbb{F}\mathcal{L}(q, \dot q, t) = (q, p, t)$: p is $\partial \mathcal{L}/\partial \dot q$ along the path, and the equivalence just proved applies. ∎

In (4.7), $\partial H/\partial q$ holds p fixed and $\partial \mathcal{L}/\partial q$ holds $\dot q$ fixed.

A first-order flow

Write $z = (q, p)$ and let $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, with $n \times n$ blocks. Hamilton's equations are

$$\dot z = J\,\nabla H(z, t), \tag{4.9}$$

with ∇ the gradient in all $2n$ variables: a first-order system with a C^1 right side, which has unique maximal solutions by Theorem 1.3 (cited), applied to (1.6). One function determines the whole vector field, a fixed matrix times a gradient; the rest of Part II studies what that implies. Along solutions,

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \frac{\partial H}{\partial q}\cdot\dot q + \frac{\partial H}{\partial p}\cdot\dot p + \frac{\partial H}{\partial t} = \frac{\partial H}{\partial t} = -\frac{\partial \mathcal{L}}{\partial t},$$

by (4.8): Theorem 3.15 again. The Hamiltonian H is conserved when $\mathcal{L}$ does not contain t, and for the Lagrangians of Lecture 2 it is $T + V$, by (3.11).

For the oscillator, $\mathcal{L} = \frac{1}{2}m\dot q^2 - \frac{1}{2}m\nu^2 q^2$ gives $p = m\dot q$, $H = p^2/(2m) + \frac{1}{2}m\nu^2 q^2$, and Hamilton's equations $\dot q = p/m$, $\dot p = -m\nu^2 q$.

For the pendulum of Lecture 2, $p = m\ell^2\dot\theta$ and $H = p^2/(2m\ell^2) - mg\ell\cos\theta$. In the units of (1.12), with the constant in the energy chosen so that the bottom has energy 0, this is $H = \frac{1}{2}p^2 + 1 - \cos q$, the form used from Lecture 5 on.

4.4 The phase-space action principle

Hamilton's equations have a variational principle of their own, in which q and p are varied independently. Lecture 8 starts its time-dependent generating functions from it. A *phase path* is a C^1 map $t \mapsto (q(t), p(t))$ on $[t_0, t_1]$ with $q(t) \in U$; its action is

$$S[q, p] = \int_{t_0}^{t_1} \Big(p\cdot\dot q - H(q, p, t)\Big)\mathrm{d}t. \tag{4.10}$$

It is *stationary with q fixed at the ends* if $\frac{\mathrm{d}}{\mathrm{d}s}S[q + s\eta, p + s\zeta]\big|_{s=0} = 0$ for all C^1 maps $\eta, \zeta : [t_0, t_1] \to \mathbb{R}^n$ with $\eta(t_0) = \eta(t_1) = 0$; nothing is asked of ζ at the ends. At $p = \partial \mathcal{L}/\partial \dot q$ the integrand is $p\cdot\dot q - H = \mathcal{L}$, by (4.6): this is the action of Lecture 3 with the momentum set free.

Theorem 4.6 (The phase-space action principle)**.** *A phase path is stationary with q fixed at the ends if and only if it satisfies Hamilton's equations* (4.8) *on* $[t_0, t_1]$.

Proof. For small s the varied path keeps q in the open set U. Differentiate under the integral and integrate $p \cdot \dot\eta$ by parts (p is C^1):

$$\frac{\mathrm{d}}{\mathrm{d}s} S\Big|_{s=0} = \int_{t_0}^{t_1} \left[\zeta \cdot \left(\dot q - \frac{\partial H}{\partial p}\right) - \eta \cdot \left(\dot p + \frac{\partial H}{\partial q}\right)\right] \mathrm{d}t + \Big[p \cdot \eta\Big]_{t_0}^{t_1}. \tag{4.11}$$

The boundary term vanishes because η does at the ends. If Hamilton's equations hold, the integral vanishes. Conversely, take $\eta = 0$ and any C^2 ζ vanishing at the ends: Lemma 3.3 gives $\dot q = \partial H/\partial p$. Then $\zeta = 0$ and any such η give $\dot p = -\partial H/\partial q$. ∎

No ζ appears in the boundary term, because $\dot p$ does not appear in the integrand. Thus p is free at the ends (Figure 4.4). Requiring $\zeta = 0$ at the ends would give the same stationary paths, since the proof used only such variations. Prescribing q at both ends gives $2n$ conditions, as many as $2n$ first-order equations can generally meet. Prescribing p as well would over-determine the motion.

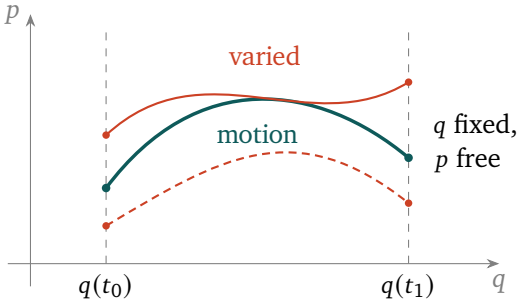


Figure 4.4: Varied paths in the phase-space action principle (Theorem 4.6): each starts on the line $q = q(t_0)$ and ends on the line $q = q(t_1)$, at any momentum. After Deriglazov, *Classical Mechanics*, Fig. 2.3.

4.5 Natural systems; Lagrange and Hamilton

Natural systems

Let $\mathcal{L} = \frac{1}{2}\dot q^{\mathsf T} M(q)\dot q - V(q)$ with $M(q)$ symmetric and positive definite (for Lecture 2's particles with fixed constraints, $M = \sum_i m_i\, Dr_i^{\mathsf T} Dr_i$, positive definite as Dr has rank n). Then $p = M(q)\dot q$ is a linear bijection in each fibre, $\mathcal{L}$ is hyperregular, and

$$H = \frac{1}{2}\, p^{\mathsf T} M(q)^{-1} p + V(q). \tag{4.12}$$

Lagrange and Hamilton

The two descriptions have the same motions wherever Theorem 4.5 applies. The table compares their uses. Entries that name a later lecture are previews of it, and nothing in this lecture depends on them.

	Lagrange (curves in Q)	**Hamilton** (flows on phase space)
Equations	n, second order	$2n$, first order, $\dot{z} = J\nabla H$
Needs	nothing to state them; W invertible to solve for $\ddot{q}$	a hyperregular $\mathcal{L}$ (here)
State	$(q, \dot{q})$; for $\mathcal{L}$ without t, orbits of states never cross, though their paths in Q may (Figure 1.1)	(q, p); for H without t, orbits of states never cross (Corollary 1.7)
Coordinates	any point transformation, and $\mathcal{L}$ is a scalar (Theorem 3.6)	point transformations lift by (4.3); also maps that mix q and p, such as $(q, p) \mapsto (p, -q)$ (Lecture 8)
Symmetry	Noether, from invariance of $\mathcal{L}$ (Theorem 3.14)	a direct test for conserved quantities (Lecture 5)
Algebra	not used in this course	a bracket operation on observables (Lecture 5)
Statistics	not used in this course	the flow preserves phase-space volume (Lecture 7)

For a system without explicit t, both state descriptions have orbits that do not cross. What the Hamilton description adds at once is that transformations may mix positions and momenta. The global change from the Lagrange description needs a hyperregular Lagrangian (Figure 4.3, and $e^{\dot{q}}$ above).

One misconception is worth naming. *"Only $\dot{p} = -\partial H/\partial q$ is dynamics."* From the Lagrangian side, $\dot{q} = \partial H/\partial p$ only restates the definition of p, as the equivalence proof in Theorem 4.5 shows, and only the second equation is Euler–Lagrange. On phase space, where q and p are independent coordinates of a state, both are equations of motion: they come from one variational principle (Theorem 4.6).

Summary.

- *Handed on:* the phase space of pairs (q, p), with p a covector (Proposition 4.1, (4.3)); for a hyperregular $\mathcal{L}$, Hamilton's equations $\dot{z} = J\nabla H$ with $H = E_{\mathcal{L}}$ in momenta (Theorem 4.5, (4.9)); natural systems $H = \frac{1}{2}p^{\mathsf{T}}M^{-1}p + V$ (4.12); the phase-space action, with q fixed and p free at the ends (Theorem 4.6).
- *Next:* Lecture 5 asks what the observables of this phase space are and how they change: the Poisson bracket. Lecture 8 starts its time-dependent generating functions from the phase-space action.

References

Arnold, *Mathematical Methods of Classical Mechanics*, §§14–15 (pp. 61–65). Cannas da Silva, *Lectures on Symplectic Geometry*, Lecture 20 (pp. 121–124). Deriglazov, *Classical Mechanics*, §2.1 (pp. 91–101; the phase paths after his Fig. 2.3). Tong, *Lectures on Classical Dynamics*, §4.1 (pp. 82–88). Fortney, *A Visual Introduction to Differential Forms*, Figs. 1.5–1.8 (pp. 14–16), for the picture of a covector.